

Intermediate Macroeconomics: Lecture 6 (Ch. 5)

ECON 311 – Intermediate Macroeconomics

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What are You Going to Learn (Ch. 5)

- Solow Growth Model
- How capital accumulates over time
- How diminishing marginal product of capital explains differences in growth rates across countries
- The principle of transition dynamics
- The limitations of capital accumulation, although a significant part of economic growth is still left unexplained

Solow Growth Model

- The Solow growth model builds on the production model
 - It provides us with a starting point to discuss why growth differs across countries
 - The model has been extended in a number of important directions
 - Foundation for modern economic growth theory (and a lot of empirical work)

- There are a few key differences:
 - Capital is endogenized: converted from an exogenous to an endogenous variable
 - Augments the production model with capital accumulation
 - Agents in the model can accumulate tools, machines, computers, and buildings over time
 - The accumulation of capital is a possible engine of long-run economic growth
 - In other words, some countries are richer than others because they invest more in accumulating capital

Setting Up the Model: Production Function

- Add an equation for the accumulation of capital over time to the production model
- The Cobb-Douglas production function:
 - Constant returns to scale
 - Increasing in both capital and labor
- Variables are time subscripted (t)

$$Y_t = F(K_t, L_t) = \bar{A}K_t^\alpha L_t^{1-\alpha}$$

The Y_t is now the output in period t , K_t is the capital in period t and L_t is the labour used in period t

Setting Up the Model: Resource Constraint

- Output can be used for consumption or investment

$$Y_t = C_t + I_t$$

C_t is the consumption in period t , I_t is the investment in period t

- Assumes no imports or exports (i.e., closed economy)

Setting Up the Model: Capital Accumulation

- Goods invested for the future determine the accumulation of capital
- Capital accumulation equation:

$$K_{t+1} = K_t + I_t - \delta K_t$$

K_{t+1} is the next year's capital

K_t is the this year's capital

I_t is this year's investment

δK_t is the depreciation of capital

δ is the depreciation rate which usually assume to be $0 < \delta < 1$

- We will use $\bar{d}$ and δ interchangeably. Typically, the estimate is around 7 to 10% annually

Setting Up the Model: Capital Accumulation

- Change in capital stock defined as

$$\Delta K_{t+1} = K_{t+1} - K_t$$

then,

$$\Delta K_{t+1} = I_t - \delta K_t$$

Future capital depends on investment today

- Today's capital is a result of investments in the past
 - This applies for all time periods in the model other than the first
 - So, we assume the economy is endowed with some initial level of capital at $t = 0$

Setting Up the Model: Labour

- For simplicity, labor supply is assumed to be inelastic and is given endogenously

$$L = \bar{L}$$

- A simple extension would be allowing for population growth, which we will cover later

Setting Up the Model: Investment

- The economy consumes a fraction of output and invests the rest

$$I_t = \bar{s} Y_t$$

where $\bar{s}$ is the fraction of output invested ($0 < \bar{s} < 1$)

- This implies

$$C_t = (1 - \bar{s}) Y_t$$

Consumption is the share of output not invested

- We will go through a model of investment and consumption later in the term

Five Equations and Five Unknowns

- Endogenous variables: Y_t, K_t, L_t, I_t, C_t
- Parameters: $\bar{A}, \bar{s}, \delta, \bar{L}, \bar{K}_0$
- Five equations:

$$Y_t = \bar{A}K_t^\alpha L_t^{1-\alpha}$$

$$\Delta K_{t+1} = I_t - \delta K_t$$

$$L_t = \bar{L}$$

$$Y_t = C_t + I_t$$

$$I_t = \bar{s}Y_t$$

- This is a dynamic model in which each equation holds at each point in time

Some more Differences between Solow and Production Model

- In the Solow model, we have added dynamics in the form of capital accumulation
- The Solow model leaves out the markets for capital and labor and the corresponding prices—the rental price of capital (r) and the wage rate (w)
 - This is a simplification that makes the model easier to solve
- The consumption equation is part of the model, but it would be redundant to include

Variables: Stock and Flow

- Stock: A quantity that survives from period to period
 - Capital in Solow model is a stock variables
 - e.g., Tractor, house, factory
 - Think of it as the saving you have in the saving amount
- Flow: A quantity that lasts a single period
 - Output, investment consumption are a flow variables
 - Think of it as the amount you just deposit to the saving account
- The change in a stock is a flow
 - Change in the capital stock equals the flow of investment

Prices and the Real Interest Rate

- We can add the wage and rental rate to our model, we would have
 - Two more endogenous variables and two more equations

$$w = MPL \text{ and } r = MPK$$

- Omitting them does not change the basic behaviour of the model

Saving in Solow Model

- National income identity also hold in the Solow model
- The saving is just the difference between income and consumption

$$\text{Saving} = Y_t - C_t = I_t = \text{Investment}$$

- Each unit of saving gets used as a unit of investment

$$\text{Saving} = \text{investment} = \text{supply of capital}$$

Solving the Solow Model

- The model needs to be solved at every point in time, which cannot be done algebraically
 - There are infinite amount of t
- Two ways to make progress:
 - Show a graphical solution
 - Solve the model in the long run
- Begin by combining equations algebraically

Solving the Solow Model

- Convert Y , K and I into per capita term

$$y_t = Y_t/L_t, k_t = K_t/L_t \text{ and } i_t = I_t/L_t$$

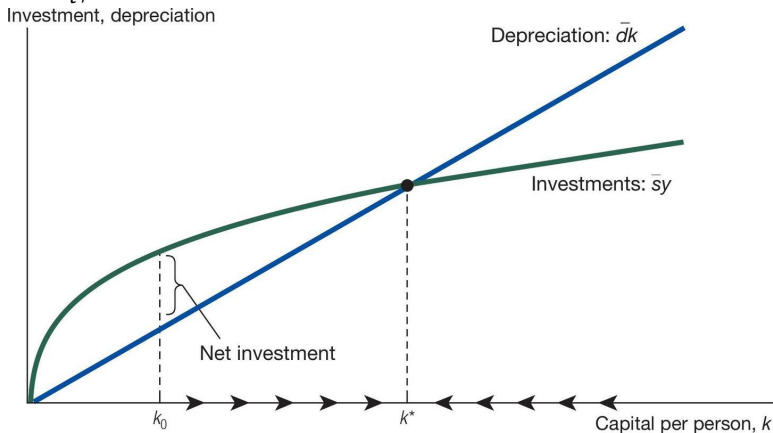
- Since we assume $L = \bar{L}$, we can solve the model at level as well (see extra question)
- Combine the investment allocation ($i_t = \bar{s}y_t$) and capital accumulation equation ($\Delta k_{t+1} = i_t - \delta k_t$)

$$\Delta k_{t+1} = \bar{s}y_t - \delta k_t$$

Change in Capital per capita = Net Investment per capita

The Solow Diagram

- Investment is $i_t = \bar{s}y_t = \bar{s}\bar{A}k_t^\alpha$
 - with the usual assumption that $0 < \alpha < 1$
 - increasing with diminishing return
- Depreciation is δK_t , which is linear



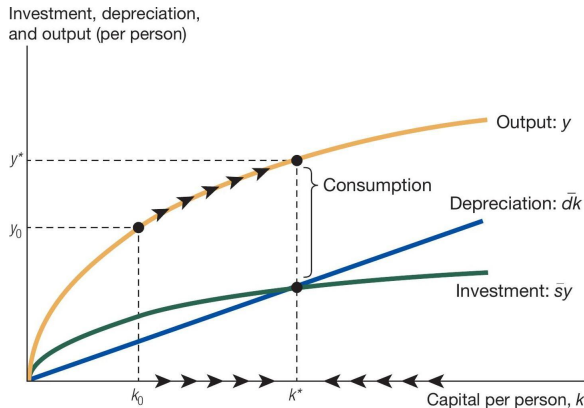
Using the Solow Diagram

- If the amount of investment $>$ depreciation, $sy_t > \delta k_t$
 - Capital stock will increase until the amount of investment being undertaken is equal to the amount of capital that wears out through depreciation $sy_t = \delta k_t$
 - This is called the steady state, capital stock will stay at this value of capital forever and the change in capital is equal to 0
- If depreciation is greater than investment, $sy_t < \delta k_t$, the economy converges to the same steady state as above
 - Capital stock will increase until the amount of investment being undertaken is equal to the amount of capital that wears out through depreciation

- When not in the steady state, the economy exhibits a change in capital toward the steady state
- As k moves to its steady state, output will also move to its steady state
- At the steady state, all endogenous variables are steady
- *Transition dynamics* take the economy from any possible initial level of capital to the steady state
- No matter what level of initial capital we choose, if we wait long enough, the economy will converge to the steady state, k^*

The Solow Diagram with Output

- We can also plot the production function to see how output evolves over time
- As transition dynamics take the economy from k_0 to k^* output rises over time from y_0 to y^*
- Consumption can also be plotted, it is the difference between output and investment



Solving for the Steady State

- We can solve mathematically for the steady-state level of capital
 - Recall that steady state is a long-run equilibrium where, if the model's variables equal a certain set of values, they will never again change from these values
- In the steady state, investment equals depreciation (i.e., $\Delta K = 0$)

$$\Delta k = 0 = \bar{s}y^* - \delta k^* \implies \bar{s}y^* = \delta k^*$$

$$y^* = \frac{\delta k}{\bar{s}}$$

Solving for the Steady State

- Substitute into the production function and solve for K^*

$$k^{*1-\alpha} = \frac{\bar{s}\bar{A}}{\delta} \implies k^* = \left(\frac{\bar{s}\bar{A}}{\delta} \right)^{1/(1-\alpha)}$$

- The steady state level of capital is positively related to the
 - investment rate ($\bar{s}$)
 - size of the workforce ($\bar{L}$)
 - productivity of the economy ($\bar{A}$)
- Negatively correlated with the depreciation rate (δ)

Solving for the Steady State

- To get y^* , plug k^* into the production function, $y^* = \bar{A}k^{*\alpha}$

- We have

$$y^* = \bar{A} \left(\left(\frac{\bar{s}\bar{A}}{\delta} \right)^{1/(1-\alpha)} \right)^\alpha \implies y^* = \bar{A} \left(\frac{\bar{s}\bar{A}}{\delta} \right)^{\alpha/(1-\alpha)}$$

$$y^* = A^{\frac{1}{1-\alpha}} \left(\frac{\bar{s}}{\delta} \right)^{\frac{\alpha}{1-\alpha}}$$

- Higher steady state production caused by higher productivity and investment rate
- Lower steady state production caused by faster depreciation

Solving for the Steady State

$$y^* = A^{\frac{1}{1-\alpha}} \left(\frac{\bar{s}}{\delta} \right)^{\frac{\alpha}{1-\alpha}}$$

- Note the exponent on productivity is different here than in the production model
 - With $\alpha = 1/3$, $\frac{1}{1-\alpha} > 1$
- Higher productivity has additional effects in the Solow model by leading the economy to accumulate more capital
 - In other word, the level of K depend on productivity
 - More output means more investment, which leads to higher levels of capital

Understanding the Steady State

- Steady state eventually reaches when we have
 - a constant level of $K^* = K_t = K_{t+1} = \dots$
 - a constant level of $Y^* = Y_t = Y_{t+1} = \dots$
 - And all other variables, $y, i, k, \dots$

Understanding the Steady State

- The economy attains a steady state because:
 - Investment exhibits diminishing returns
 - The growth rate of production and investment diminishes as the capital stock grows
- Additionally, a constant fraction of the capital stock depreciates every period, with this depreciation rate remaining constant even as the capital accumulates
 - Depreciation is not subject to diminishing returns as capital increases
- Consequently, net investment eventually becomes zero, marking the point at which the economy enters a steady state.

Economic Growth in the Solow Model

- Important result: there is still no long-run growth in the Solow model
- In the steady state, growth stop, and variables, total output (Y), capital stock(K), output per Capita (y), per capita consumption (c) and so forth remain constant
- The economy grows (or shrink) from k_0 to k^* , but eventually growth stops as the capital stock and production converge to constant levels

Economic Growth in the Solow Model

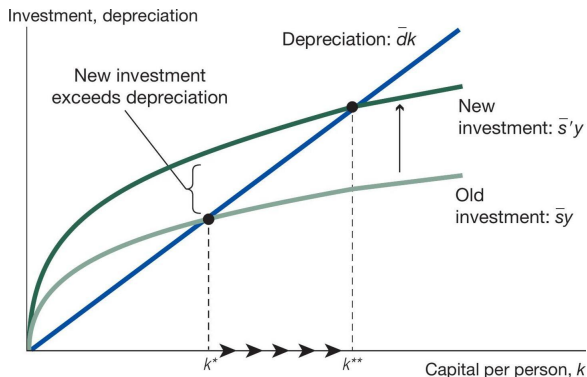
- Economies empirically exhibit continuous growth
- The model shows a limitation in explaining this growth
- According to the model, capital accumulation isn't the source of long-term economic growth
 - Savings and investment matter for growth in the medium run, but
 - In the long run, diminishing returns to capital accumulation cause the return on these investments to fall
 - However, they don't support long-term growth due to diminishing returns
- A possible explanation for this could be that we have not reached the steady state... but most economist would agree that there is something else that is driving the growth

Some Economic Experiments

- The Solow model
 - does not explain long-run economic growth
 - does help explain some differences across countries
- Economists can experiment with the model by changing parameter values

An Increase in the Investment Rate

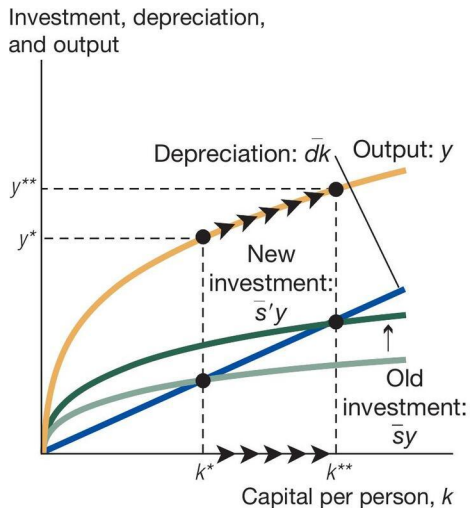
- Suppose the investment rate increases permanently for exogenous reasons ($s' > s$)
 - The investment curve rotates upward ($\bar{s}'\bar{A}k_t^\alpha > \bar{s}\bar{A}k_t^\alpha$)
 - The depreciation curve remains unchanged (δj_t)
 - The capital stock increases by transition dynamics to reach the new steady state
 - This happens because investment exceeds depreciation ($sy_t > \delta k_t$)
- The new steady state is located to the right



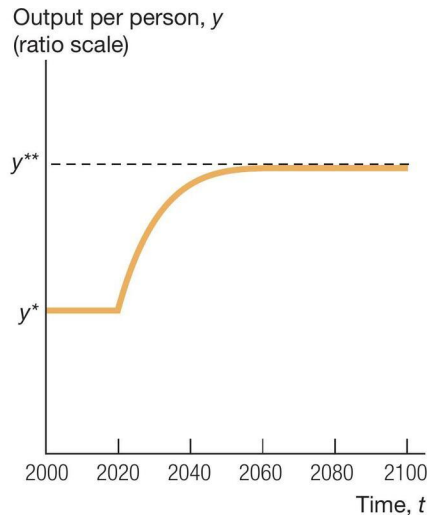
An Increase in the Investment Rate

- What happens to output in response to this increase in the investment rate?
 - The rise in investment leads capital to accumulate over time
 - This higher capital causes output to rise as well
 - Output increases from its initial steady state level y^* to the new steady state y^{**}
 - The increase in the investment rate causes the economy to grow over time, until a new steady state is reached
 - In the long run, both steady state capital and steady state production are higher
 - Because labour is constant, this also means the level of output per person is permanently higher as well

The Behaviour of Output after an Increase in $\bar{s}$



(a) The Solow diagram with output.



(b) Output per person over time.

Next Lecture

- Next lecture will finish Ch.5